



Helping Hands

Kalyan Varma Alluri: Linux networking guru, has been with Yahoo! Inc for 3 years. A PESIT Alumnus and a photographer as well, he helped get sponsorship from Yahoo!.

Babu Kalakrishnan: Ex-scientist, ISRO, helped in circuit-design and communication issues. Was the unofficial project guide.

Atul Chitnis: Helped procure the Simputer as well as sonar modules and electronic compasses during a crisis.

COMRADE. Born as a set of simple 2D-image processing operations, IRIS evolved into an optimised, object-based, machine-vision system capable of complex vision tasks. “IRIS is a system that allows robots to see and base their planning and decision upon visual feedback. It is different from other image-processing software because it uses advanced algorithms to achieve functionality and implementation on a case-to-case basis.”

The IRIS Engine is not limited to COMRADE. Any other robotic system can use it as well. Though mainly working from a command-line interface, it provides a real-time OpenGL plot of the robots’ location. This node transmits the collective tasks for Adam and Eve as BorgCode, which is transmitted through WiFi to the robots.

A robot cannot execute all the instructions present in the BorgCode. For example, Eve cannot execute an instruction to visually search for an object since it does not have a Webcam.

Thus, a “decompositor” breaks down BorgCode into executable chunks of code, each of which calls for a single feature of the robot; it is these chunks that are downloaded onto the individual robots for execution.

“We were forced to simplify it. The original idea was two to three robots interacting,” says Sharad.

The four underwent a nightmare debugging the code. Sharad says that it was a great experience, but it didn’t quite go where they wanted it to go. Pressured by PESIT to finish the project on time, they compromised on a lot of things.

The autonomous robots are in place and working, but the four agree that the co-operation part needs to be worked upon. The robots work well by themselves, but they need to be “married,” to fulfil the project goal.

Abhishek wants to pursue artificial intelligence, and take IRIS forward. And Sharad is leaving for the University of Edinburgh. “Yeah, it feels bad that he is leaving,” muses Mrinal, “but you see... it was a college project. And now that all of us are working, it has become a hobby.” Sharad echoes his thoughts.

“Once Sharad leaves, we might start working again,” says Das, uncertainly. “But it requires a team, you know.” Sharad agrees: “Most of the work happened because we were a team. Once that breaks...” he trails away.

Das and Sen Gupta are colleagues now, and Mrinal shares Bangalore with them. He stays in Banshankari, the other two in Koramangala.

Das reminisces how Adam once broke down, and Sharad adds, “It was a nightmare, I mean, a motor short circuits, and you don’t know what to do.”

Meanwhile, Adam and Eve wait for their final union. “We thought some of our juniors would be interested in our project. But no one seems to be ready,” says Sharad.

His first love was Digital Signal Processing, and he intends to go back to it in the future. The comrades have decided to disband.

The revolution remains unfinished. ☒

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